CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 12: **How Nature Works in Harmony** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Plants lose water vapour through tiny pores called: 1

(a) nodules **(b)** stomata **(c)** petals **(d)** roots

2. A rise in water temperature of a lake would cause: 1

(a) more gases **(b)** more oxygen
(c) less dissolved oxygen **(d)** no effect

3. The greatest number of individuals in a food chain is usually of: 1

(a) producers **(b)** top carnivores **(c)** herbivores **(d)** carnivores

4. Which process returns CO2 to the air? 1

(a) photosynthesis **(b)** fertilisation **(c)** respiration **(d)** pollination

5. Overgrazing leads to: 1

(a) more forests **(b)** more wildlife
(c) loss of plant cover and soil erosion **(d)** more rain

6. Which of the following would be a secondary consumer? 1

(a) a cow eating grass **(b)** a frog eating insects
(c) a decomposer **(d)** a plant

7. Legume plants enrich the soil with: 1

(a) sulphur only **(b)** phosphorus only
(c) nitrogen compounds **(d)** carbon

8. Which action is MOST sustainable? 1

(a) dumping waste **(b)** cutting forests
(c) burning plastic **(d)** using biodegradable material

9. Which of the following statements about decomposers is true? 1

(a) They are not needed **(b)** They recycle nutrients to the soil
(c) They produce food by photosynthesis **(d)** They eat only green plants

10. Crop rotation with legumes helps to: 1

(a) reduce rain **(b)** restore soil nitrogen
(c) increase salt **(d)** kill all insects

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. State two ways in which fungi and bacteria help the farmers. 2

12. Why do we plant trees along roads? 2

13. Why do aquarium owners keep green plants in the tank? 2

14. How do large grass areas help to reduce soil erosion? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student says 'decomposers are the enemies of living things.' Respond using three points. 3

16. What do you mean by an endangered species? Give two examples and a conservation step. 3

17. State three effects of the excessive use of chemical fertilisers. 3

OR

Explain how eating lower in the food chain can feed more people. 3

18. Explain how acid rain harms ecosystems, naming the causes and effects. 3

OR

Explain two ways in which animals depend upon plants and two ways in which plants depend upon animals. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the carbon and oxygen balance in the biosphere including photosynthesis, respiration, decomposition and burning. What happens if deforestation continues? 5

OR

Discuss with examples the importance of conservation of forests, wetlands and oceans for humans and other organisms. 5

20. Explain the nitrogen cycle with the processes of fixation, absorption, passage along food chains, and return, and the role of legumes in farming. 5

OR

A town dumps garbage in a nearby lake. Predict the changes in the lake ecosystem in order and suggest remedies. 5

END OF QUESTION PAPER — Total: 40 Marks